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RESEARCH ARTICLE

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Effectiveness of *Scutellaria barbata* water extract on inhibiting colon tumor growth and metastasis in tumor-bearing mice

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The plant *Scutellaria barbata* (SB) is commonly used as herbal medicines for treating cancer. The present pre-clinical study aimed to validate the Chinese Pharmacopoeia (CP) recommended dosages of SB water extract (SBW) in treating colon tumors. The content of chemical marker scutellarin in SBW was quantified using UPLC. Mice bearing human HCT116 xenografts or murine colon26 tumors received oral administration of SBW or scutellarin for 4 weeks. Results showed that SBW (615 and 1,230 mg/kg) and scutellarin (7 mg/kg) treatments significantly reduced human xenograft weights by 28.7, 36.9 and 28.8%, respectively. Lung metastasis area could be ameliorated after SBW (615 mg/kg) and scutellarin (7 mg/kg) treatments by 23.4 and 29.5%, respectively. Expressions of colon cancer metastasis-related proteins E-cadherin, Tspan 8 and CXCR4, as well as Src kinase in tumors were first shown to be regulated by SBW. Furthermore, in murine colon26 tumor-bearing mice, SBW (615 mg/kg) and scutellarin (7 mg/kg) treatments reduced the orthotopic tumor burden by 94.7% and lung metastatic tumor burden by 94.1%, respectively. Our findings provided evidences that SBW (at the mouse equivalent dosages to clinical dosages recommended by CP) could exert anti-tumor and anti-metastatic effects in colon cancer animal models.

KEYWORDS
chemical marker, colon cancer, metastasis, *Scutellaria barbata*, scutellarin

1 | INTRODUCTION

Scutellariae Barbatae Herba (SB) is the dried whole plant of *Scutellaria barbata* D. Don, which is a perennial plant belonging to the Labiaceae family and is widely distributed in northeast Asia. As recorded in Chinese Pharmacopoeia 2015 (CP 2015), Scutellariae Barbatae Herba can be used for clearing heat, removing toxin, resolving stasis and promoting urination (Chinese Pharmacopoeia Commission, 2015). SB was used in combination with other herbs to treat different types of hemorrhoids, carbuncles or tumors (Chen, 1993). Furthermore, SB has long been used as an important component in several TCM formulae to treat various kinds of cancer (Hsu, 1982; Q. Wang, Acharya, et al., 2018) and a survey showed that treating cancer is the most common use of SB by TCM practitioners (Tao & Balunas, 2016).

In modern pharmacological studies, SB is recognized as a Chinese medicinal herb that can modulate mutagenesis, possess anti-carcinogenic effects and inhibit murine renal cell carcinoma growth by a research group in the United States in 1990s (Wong, Lau, Jia, & Wan, 1996). In the later studies, SBW was shown to inhibit the proliferation of myometrial and leiomyomals (Lee, Kim, Han, & Kim, 2004) and induce apoptosis of hepatoma H22 cells (Dai et al., 2008). A Phase I clinical trial, BZL101 (an aqueous extract of SB) was demonstrated to be safe and have a favorable toxicity profile in breast cancer patients (Rugo et al., 2007). While the inhibitory effects of SB ethanol/methanol extract on human lung (Z. Wang, Qi, et al., 2018; Yin,

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Zhou, Jie, Xing, & Zhang, 2004) and colon (Goh, Lee, & Ong, 2005; L. Wei et al., 2013) cancer cells were also reported.

On the other hand, the in vivo efficacies of SB extracts have also been evaluated in prostate (Wong et al., 2009), liver (Dai et al., 2011; Kan et al., 2017) and colorectal (J. Lin et al., 2014; L. Wei et al., 2012) cancer mice models and promising anti-tumor effects could be observed. Nevertheless, the effective dosages of SB extracts in these animal models might not be necessarily translatable to human dosages. The anti-tumor effects observed in mice feeding either with SB ethanol extract (Kan et al., 2017; J. Lin et al., 2014; L. Wei et al., 2012), which is not the common practice to consume SB, or with relatively high dosages of SBW (Dai et al., 2011; Wong et al., 2009). As the daily dosages of SB recommended in CP 2015 are 15–30 g, the efficacy of SBW at these dosages which have not yet been validated in any preclinical study, would be investigated in our present study.

In view of the fact that SB is one of the most commonly prescribed single herb for colon cancer patients and survivors (Chao et al., 2014; Chen, Zhao, & Cong, 2018; T. H. Lin, Yen, et al., 2017), and the traditional use of SB in treating carbuncle in intestine (Hsu, 1982), the efficacies of SBW as well as the chemical marker of SB, scutellarin, were evaluated and compared using mice bearing human colon xenografts or murine colon tumors. Here, the anti-tumor and anti-metastatic activities of SBW (at the actual clinical dosages) in colon cancer mice models were first demonstrated and the involved mechanisms were further elucidated in this study.

2 | MATERIALS AND METHODS

2.1 | Herbal materials and extraction

The raw herb of SB was obtained from a renowned herbal supplier in Hong Kong. Morphological and chemical authentication of raw herb was performed as described in CP 2015 (Chinese Pharmacopoeia Commission, 2015). Authenticated voucher specimen (No. 3503) was deposited in the museum of the Institute of Chinese Medicine, The Chinese University of Hong Kong. Chemical profiles of the raw herb and reference marker, scutellarin (CAS No.: 27740-01-8, obtained from Chengdu Must Bio-Technology Co. Ltd., Chengdu, China) were compared using thin layer chromatography. Dried SB (1 kg) were meshed into powder form and then immersed in distilled water (10 L) for 1 hr, followed by reflux extraction with boiling water for 1 hr and then repeated once. The combined water extracts were filtered and concentrated under vacuum, followed by lyophilization (Tokyo Riakikai, Tokyo, Japan) in a freeze-drier (Labconoco, MO) and finally became powder form. The SBW powder was stored in desiccator at room temperature. The percentage yield of SBW was 20.26% (wt/wt). SBW was subjected to standardization in which the chemical profile was registered (Figure 1) and chemical marker scutellarin was determined quantitatively using UPLC.

SBW was dissolved in cell culture medium and distilled water for in vitro and in vivo studies, respectively. Scutellarin (Scu) was dissolved in DMSO and added in medium or water (<0.5% vol/vol DMSO) for experiments.

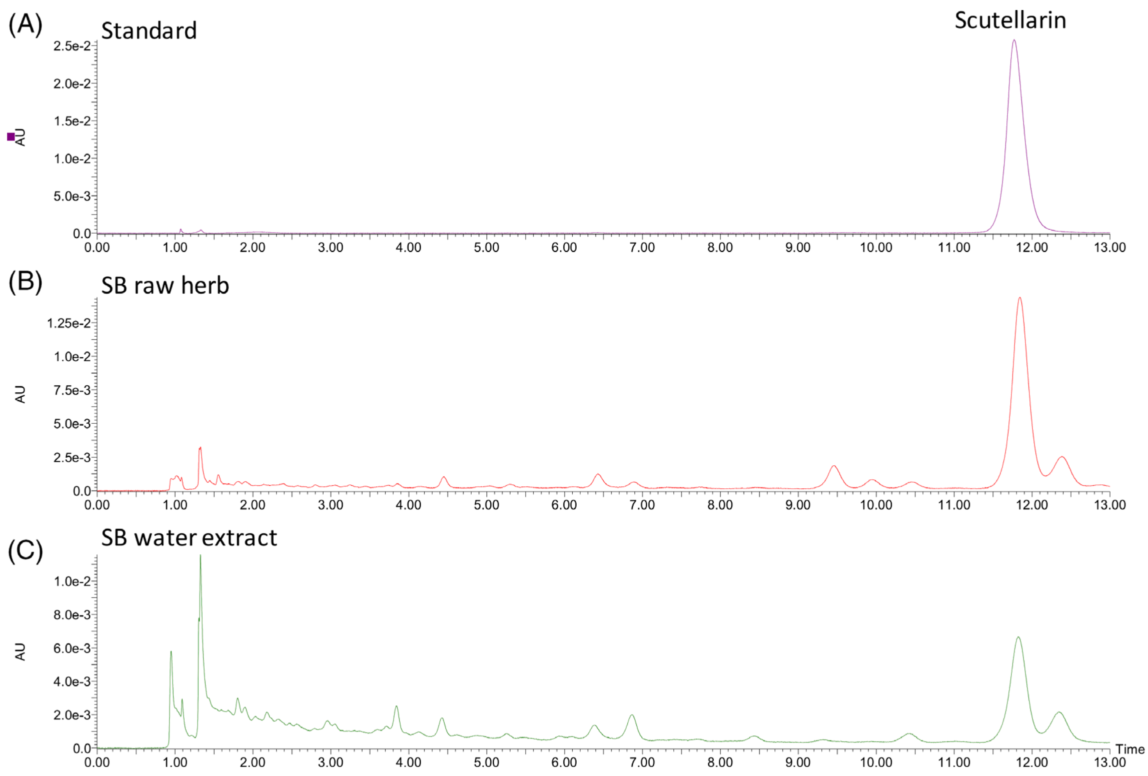

FIGURE 1 The representative UPLC chromatograms of (a) scutellarin standard, (b) SB raw herb and (c) SB water extract. SB, *Scutellaria barbata* [Colour figure can be viewed at wileyonlinelibrary.com]

2.2 | UPLC analysis of SB crude extract

Stock solution of scutellarin (Scu) was prepared in methanol at 1 mg/ml and was stored at −20°C until use. The stock standard solution was serially diluted into 100, 50, 25, 12.5, 6.25, 3.125 μl/ml as the standard curve solution. One gram of powder sample was weighed accurately to a Soxhlet extractor. Petroleum ether (60–90°C) was added and heated under reflux until the extract became colorless. The petroleum ether solution was discarded and the petroleum ether of the residue was expelled. Methanol was added afterwards for further heating under reflux until the methanol extract became colorless. The extract was transferred to a 100 ml volumetric flask and methanol was added to volume. The above solution (25 ml) was evaporated to dryness and the residue was dissolved in 20% methanol and diluted to 25 ml and mixed. The solution was then filtered as the test solution.

The UPLC analysis was conducted using Waters ACQUITY UPLC (Waters, MA). The column used was Agilent Poroshell C18 (2.6 μm, 3 × 100 mm) with Agilent Poroshell C18 (2.6 μm, 3 × 10 mm) as guard column (Agilent Technologies, CA). The chromatographic separation was conducted at 40°C under isocratic elution of 0.1% acetic acid in deionized water: acetonitrile (88:12) at a flow rate of 1 ml/min. UV335 nm was used for the detection of scutellarin. The percentage yield of Scu in SBW was 0.538% (wt/wt).

2.3 | Cell culture and cell viability assay

Human colon cancer cell line HCT116 and human skin fibroblast Hs27 was purchased from American Type Culture Collection (VA). Cell culture reagents were all purchased from Life Technologies (CA). Colon26-Luc cells (JCRB1496) were purchased from Japanese Collection of Research Bioresources Cell Bank (Osaka, Japan). HCT116 cells were grown in McCoy 5A medium and Hs27 cells were grown in DMEM medium, both of these media were supplemented with 10% (vol/vol) fetal bovine serum (FBS) and 1% (vol/vol) penicillin–streptomycin, while Colon26-Luc cells were grown in RPMI 1640 medium supplemented with 2 mM L-glutamine and 10% (vol/vol) FBS and 1% (vol/vol) penicillin–streptomycin. All culture media and supplements were obtain from Thermo Fisher Scientific (MA). Cells were maintained in culture flasks in a humidified incubator with 5% CO_2 at 37°C. Cells were detached by trypsinization with trypsin-EDTA (0.25%) when they reached 80% confluence and subcultured every 3–4 days.

HCT116 cells or Hs27 cells (5×10^3/well) were seeded into 96-well plates and precultured for 24 hr, then treated with different concentrations of SBW (50–800 μg/ml) and Scu (9.75–156 μM) for 48 hr. The tested concentrations of Scu were based on a previously reported IC_{50} value of Scu (78 μM) in HCT116 cells (R. J. Jiang et al., 2013). The tested concentration range of SBW was applied according to our previous studies (G. G. L. Yue et al., 2013, 2015) as no reference IC_{50} value in the same cell line is available. Cell viability was determined by MTT assay (G. G. L. Yue et al., 2015).

2.4 | Cell migration and invasion assays

Transwell inserts (6.5 mm, Costar, VA) containing polycarbonate filters with 8 μm pores were used for migration and invasion studies. For migration assay, HCT116 cells (2×10^4/well) were seeded with different concentrations of SBW and Scu, and were put into apical chambers containing 1% FBS medium, whereas 10% FBS medium was added into basolateral chambers. After 11 hr incubation, cells were fixed with methanol and stained with hematoxylin. Cells remained in apical chamber were removed by wet cotton swabs. Each bottom layer of transwell membranes was photographed for four different fields with a light microscope for cell counting.

In the invasion assay, transwell inserts were coated with growth factor reduced (GFR) matrigel (BD Biosciences, CA) and incubated at 37°C for 30 min until gelification, prior to cell seeding. Cells were then seeded with different concentrations of SBW and Scu, and were put into apical chambers and then incubated for 24 hr to allow invasion into basolateral chamber. The cell counting procedures were the same as those in migration assay.

2.5 | Human colon xenograft-bearing mouse model

Male nude mice (6–8 weeks of age) were inoculated subcutaneously with HCT116 cells (5×10^6 per mouse) on the back. Treatments were started when the tumor volume reached 50 mm^3. Dosages of SBW and Scu for animal studies were calculated in accordance to CP 2015. Since the recommended dosages of SB in CP 2015 are 15–30 g, and the extraction yield was 20.26%, the maximum human dosage of SB would be 101 mg/kg. The dosages for mice would be 1,230 mg/kg according to the US FDA guidance "Conversion of Animal Doses to Human Equivalent Doses" (U.S. Department of Health and Human Services). Hence, the dosages of SBW at 615 and 1,230 mg/kg were used. While the content of Scu in SBW was 0.538% (wt/wt), therefore, two dosages of Scu at 3.5 and 7 mg/kg were used. Mice bearing HCT116 tumors were orally administered with SBW at 615 mg/kg (SBW-L), 1,230 mg/kg (SBW-H), scutellarin at 3.5 mg/kg (Scu-L) or 7 mg/kg (Scu-H) for 4 weeks. Tumor sizes were measured twice a week. Tumor volumes were calculated using formula [(length × width × depth)/2 mm^3] as described previously (M. Y. Li et al., 2017). At the end of experiment, mice were anesthetized and whole blood were collected for plasma enzyme levels determination. Then the mice were sacrificed by cervical dislocation, then tumors and lungs were collected for further analysis. Concentrations of alanine aminotransferase (ALT), aspartate aminotransferase (AST) and lactate dehydrogenase (LDH) and creatine kinase (CK) in plasma were analyzed using respective enzyme assay kits according to the manufacturer's instructions (Stanbio Laboratory, TX).

2.6 | Orthotopic colon tumor-bearing mouse model

Male BALB/c mice (6–8 weeks of age) were inoculated with luciferase-tagged mouse colon tumor cells (Colon26-Luc) at 1×10^6 cells per

40 μl phosphate buffered saline. Cells were injected into the posterior wall of rectum in mouse under anesthesia. Successful cell inoculation was confirmed by live animal whole-body imaging using IVIS 200 Imaging system (Xenogen, Alameda, CA) 3 days after cell inoculation. Images were acquired for 1–60 s and the photon emission transmitted from mice were quantified using Xenogen Living Image (Igor Pro version 3.2 software), and graphed according to the average radiance (photons/s/cm^2/sr) (Ko et al., 2017). The treatments of SBW and Scu started and the protocol were the same as in human colon xenograft-bearing mice mentioned above. During the treatment, IVIS imaging was performed weekly and on the day of sacrifice in order to monitor the growth of orthotopic tumors. At the end of experiment, mice were anesthetized, undergone IVIS imaging and blood collection as mentioned in Section 2.5 for plasma enzymes determination. Then the mice were sacrificed and lungs were collected for IVIS imaging, whereas spleens and tumors were collected for immune cells population determination as described previously (M. Li, Yue, Tsui, Fung, & Lau, 2018).

Nude mice and BALB/c mice used in these models were provided by Laboratory Animal Services Center of The Chinese University of Hong Kong and were kept under pathogen-free conditions. All experiments were approved by the Animal Experimentation Ethics Committee of The Chinese University of Hong Kong (Ref. No. 14/170/MIS and 18/231/MIS).

2.7 | Histological analysis of lungs

Lungs of nude mice were fixed in 10% formalin and processed for embedding in cassettes. Tissues were sectioned longitudinally at 5 μm thickness. Lung sections were collected on gelatin-coated slides for hematoxylin & eosin staining. Stained slides were photographed with a light microscope (Olympus IX71, Olympus, Tokyo, Japan). Then the blinded assessment of metastasis area in lung photos were performed by two researchers using Image J software (NIH, MD). Tumor burden were presented as the average percentage of tumor areas within organ areas.

2.8 | Western blotting of tumors

HCT116 tumors from nude mice were first homogenized in 1 × PBS for cell extraction. Cells were then lysed with lysis buffer (Beyotime Institute of Biotechnology, Shanghai, China) on ice. After that, cell lysates were centrifuged and supernatant were collected and stored at −80°C until use. Protein concentration was determined by a BCA kit (Thermo Fisher Scientific). Before gel electrophoresis, protein samples were heated at 95°C. Equivalent amount of proteins were loaded at 10% SDS-PAGE gels for 2 hr at 100 V. Proteins on gel were transferred to polyvinylidene fluoride (PVDF) membranes (Immobilon, Merck KGaA, Darmstadt, Germany) electrophoretically for 1–1.5 hr at 90 V. The membranes were blocked with 5% non-fat milk (wt/vol) in Tris-buffered saline Tween 20 (TBST) and then incubated with primary antibodies (1:1000) on a shaker at 4°C overnight. After incubation, membranes were washed with TBST and incubated with secondary horseradish peroxidase-conjugated antibodies for 1 hr on a shaker. Protein bands were detected by enhanced chemiluminescence (ECL) solution (GE Healthcare, IL) and were captured by a molecular imager, ChemiDoc XRS+ (Bio-Rad, CA). Signals were quantified by Image J software and normalized by the housekeeping proteins (β-actin or GAPDH). Quantitative data were represented as fold changes of untreated control.

2.9 | Flow cytometric detection of MDSC populations

Cells were harvested from spleens and tumors of BALB/c mice for population determination of myeloid-derived suppressor cells (MDSC). The excised spleens and tumors were pressed through the nylon membranes using syringe plungers and the isolated cells were washed and blocked using PBS supplemented with 5% FBS. Cells were stained with a cocktail of PE-anti-mouse CD 11b and FITC-anti-mouse Ly-6G (Gr-1) (BD Biosciences). All stained cells were analyzed on a FACSCanto™ II Flow Cytometer (BD Biosciences). Data from 10,000 cells were collected and analyzed.

2.10 | Statistical analysis

All data were analyzed by GraphPad Prism 8.0 software package (GraphPad Software Inc., San Diego, CA). in vitro results were presented as mean ± *SD*, while in vivo results were presented as mean ± *SEM*. One way analysis of variance (ANOVA) with Dunnett test or Tukey's *post hoc* tests were used to analyze the quantitative results. In all comparison, statistical significance was considered when $p < .05$.

3 | RESULTS

3.1 | SBW and Scu decreased human colon cancer cell viability

After 48 hr incubation, both SBW and Scu significantly decreased the viability of HCT116 cells in a concentration manner (Figure 2A,B). The IC_{50} value of SBW was 271.9 μg/ml and that of Scu was 105.4 μM (equivalent to 48.7 μg/ml). With the known content of Scu present in SBW (0.538%), the inhibition curve of SBW was converted to the corresponding Scu content and then compared with the inhibition curve of Scu alone. As shown in Figure 2C, lower IC_{50} value could be obtained on the inhibition curve of SBW (1.5 μg/ml) than that of Scu alone (48.7 μg/ml), showing that at the same concentration of Scu, the inhibitory potency of SBW was higher than that of Scu alone in HCT116 cells, implying that other active components are present in SBW. On the other hand, when the human fibroblasts Hs27 were

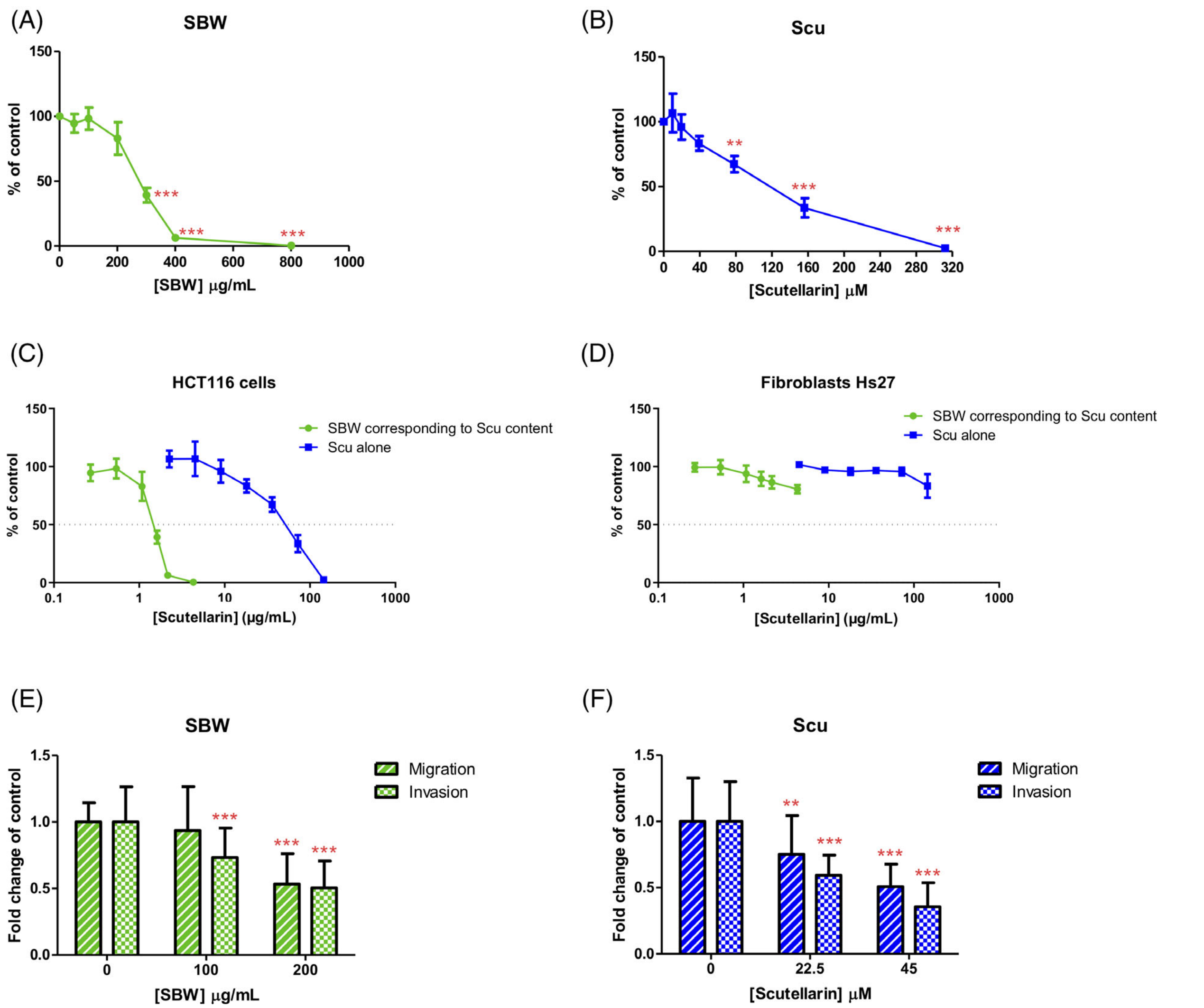

FIGURE 2 Effects of SB water extract (SBW) and scutellarin (Scu) on human HCT116 colon cancer cells. Cell viability of cells treated with (a) SBW or (b) Scu were assessed using MTT assay. Data were expressed as the mean percentage of the untreated control (3 independent experiments with 5 replicates each, mean ± *SD*). Inhibition curves of SBW and Scu on (c) HCT116 cells and (d) Hs27 cells viability were plotted against the concentration of Scu for comparison. In migration assay, cells were treated with (e) SBW or (f) Scu at non-cytotoxic concentrations in the transwell with or without matrigel coating. Data were expressed as the mean fold change of untreated control (4 independent experiments with duplicate each, mean + *SD*). Statistical differences were determined by one-way ANOVA, followed by Dunnett test, with ***p* < .01, ****p* < .001 against untreated control. ANOVA, analysis of variance [Colour figure can be viewed at wileyonlinelibrary.com]

incubated with SBW and Scu, they did not cause high cytotoxicity (Figure 2D) whereas they significantly decreased viability in HCT116 cancer cells, suggesting the selective cytotoxic activities of SBW and Scu.

3.2 | SBW and Scu inhibited human colon cancer cell migration and invasion

As cell migration and invasion play key roles in cancer metastasis, the inhibitory effects of SBW and Scu on migration and invasion of HCT116 cells were explored. Non-toxic doses of SBW and Scu were separately added into HCT116 cells in transwell migration and invasion assays. Results showed that both SBW (200 μg/ml) and Scu (22.5–45 μM) significantly attenuated migration through the transwell membrane on HCT116 cells in a dose dependent manner (*p* < .001, Figure 2E). Likewise, both SBW (100–200 μg/ml) and Scu (22.5–45 μM) significantly decreased the number of invaded cells in matrigel-coated transwells (Figure 2F).

3.3 | SBW suppressed tumor growth and metastasis in HCT116 xenograft mice

The in vivo efficacies of SBW and Scu were further investigated in HCT116 xenograft-bearing nude mice. Mice were treated with SBW at CP 2015 recommended daily human equivalent dosages, that is, 615 mg/kg (SBW-L) and 1,230 mg/kg (SBW-H) for 4 weeks.

The dosages of scutellarin at 3.5 mg/kg (Scu-L) or 7 mg/kg (Scu-H) were calculated basing on the SBW daily doses and the percentage yield of scutellarin in SBW. After 4 weeks treatments, body weight of the mice had no significant changes among all groups (Figure 3A). During the treatment period, from Day 19, tumor volume of SBW-H treatment group was shown to be lower than that of untreated control group ($p < .05$, Figure 3B), while SBW-L and Scu-H treatments also resulted in significant decreases on the final tumor volume. Furthermore, the final tumor weights of SBW-treated groups were significantly lower than that of untreated control group ($p < .05$, Figure 3C). The inhibition rates of SBW-L and SBW-H were 28.7 ± 6.9% and 36.9 ± 6.0%, respectively. Besides, Scu-H treatment also resulted in significant decrease in tumor weight (inhibition rate of 28.8 ± 7.5%), while Scu-L treatment could only slightly suppress (without significant difference against untreated control group). As the amount of Scu in SBW-L and Scu-L were the same, other components present in SBW might also contribute to the inhibitory activities of SBW on tumor growth, which is consistent with our in vitro results (Section 3.1).

To further estimate whether SBW and Scu treatments induce toxicity in mice, levels of liver- and muscle-specific enzymes in plasma were assessed. Results showed that the no significant difference in the levels of liver-specific plasma enzymes ALT, AST and LDH and muscle-specific enzyme CK among treatment groups (Figure 3D), suggesting that SBW and Scu treatments did not cause toxic effects in xenograft-bearing mice.

Since the inhibitory potency of SBW-L was comparable to that of Scu-H, the tumor burden in lungs as well as the tumor growth regulatory proteins expression levels in tumors were further analyzed among these two treatment groups and untreated control group.

As shown in microscopic photos, tumor nodules (yellow arrows in Figure 3E) could be seen in the lung sections of HCT116 xenograft-bearing mice. The metastasis area in lungs were assessed and the quantitative results were shown in Figure 3F. Both SBW-L and Scu-H treatments slightly reduced the metastasis area in lungs by 23.4 ± 11.3% and 29.5 ± 12.6%, respectively, though the differences among groups were not statistically significant.

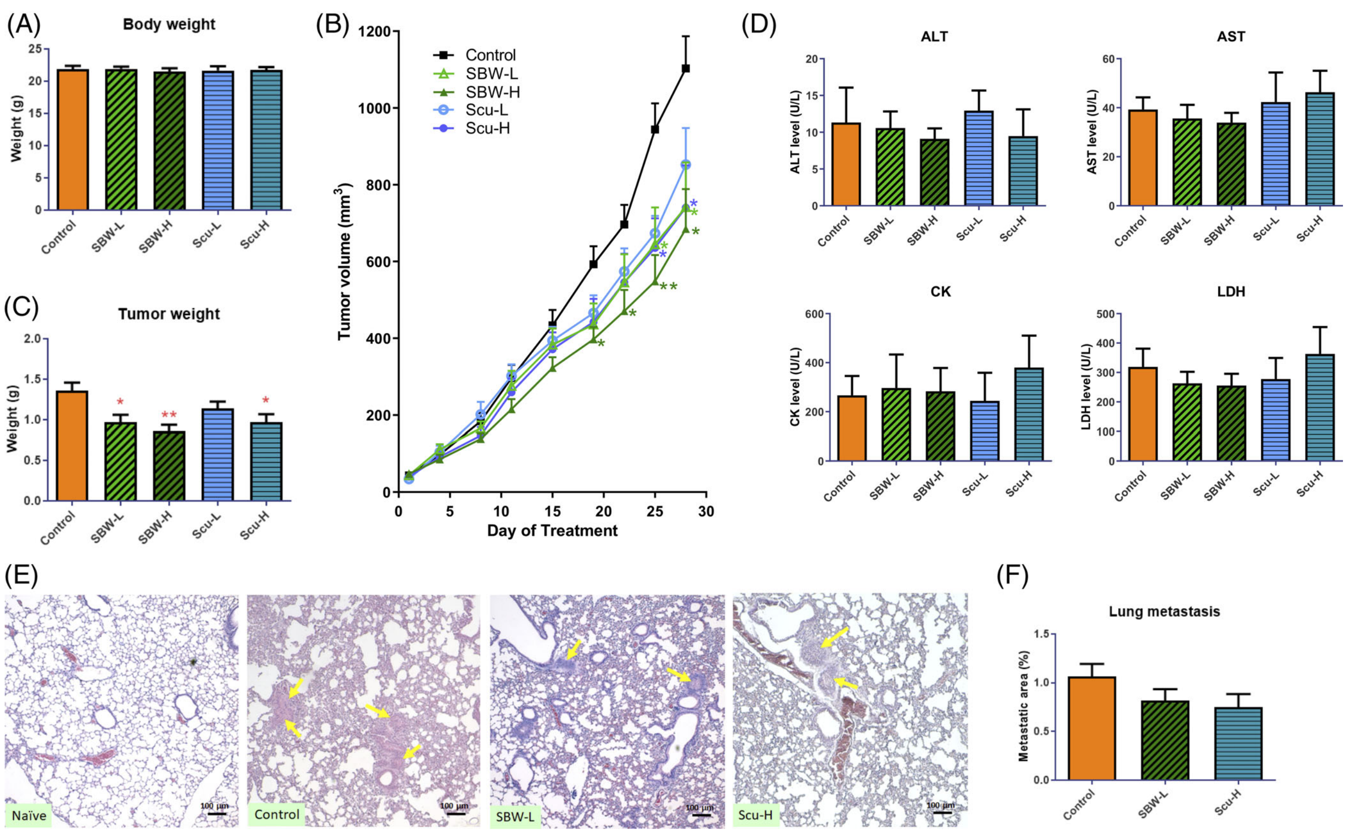

FIGURE 3 Effects of SB water extract (SBW) and scutellarin (Scu) treatments on colon tumor-bearing mice. Mice were treated with SBW-L (615 mg/kg), SBW-H (1,230 mg/kg), Scu-L (3.5 mg/kg) or Scu-H (7 mg/kg) daily for 4 weeks. (a) Body and (c) tumor weights of mice after treatments were recorded at the end of treatment. (b) Tumor volume were recorded twice a week during treatment and tumor growth curves were shown. (d) Effects of SBW and Scu treatments on the levels of plasma enzymes ALT, AST, LDH and CK in HCT116 tumor-bearing nude mice after treatment. (e–f) Effects of SBW and Scu treatments on lung metastasis. Lungs were collected from naive mice (without tumor) and colon tumor-bearing mice. Sections of lungs were stained with H&E and photographed. (e) Representative photos from Control, SBW-L, Scu-H group showed tumor nodules (yellow arrows). (f) Tumor burden were presented as the mean percentage of metastatic area. Mean + *SEM*, $n = 10–12$ in each group. Differences among treatment groups were determined by one-way ANOVA followed by Tukey's multiple comparisons test, $^{*}p < .05$, $^{**}p < .01$ as compared with control group. ANOVA, analysis of variance [Colour figure can be viewed at wileyonlinelibrary.com]

3.4 | SBW regulated expressions of the growth- and metastasis-related proteins in HCT116 tumors

To determine the effects of SBW and Scu on the progression and metastasis potential of tumors, proteins were extracted from tumors of SBW-L and Scu-H treated groups and untreated control group. From the western blot results, SBW-L and Scu-H treatments enhanced the expressions of cleaved caspase-3 and cleaved PARP in the tumors (Figure 4A,B). Furthermore, the expression of p53, which has been previously reported to be affected by scutellarin in colon cancer cells (Yang, Zhao, Wang, Liu, & Zhang, 2017), has also been shown to increase in the tumors of mice after SBW-L and Scu-H treatments (Figure 4C,D).

On the other hand, as the reduced lung metastasis was observed in SBW-L- or Scu-H-treated mice, the expressions of

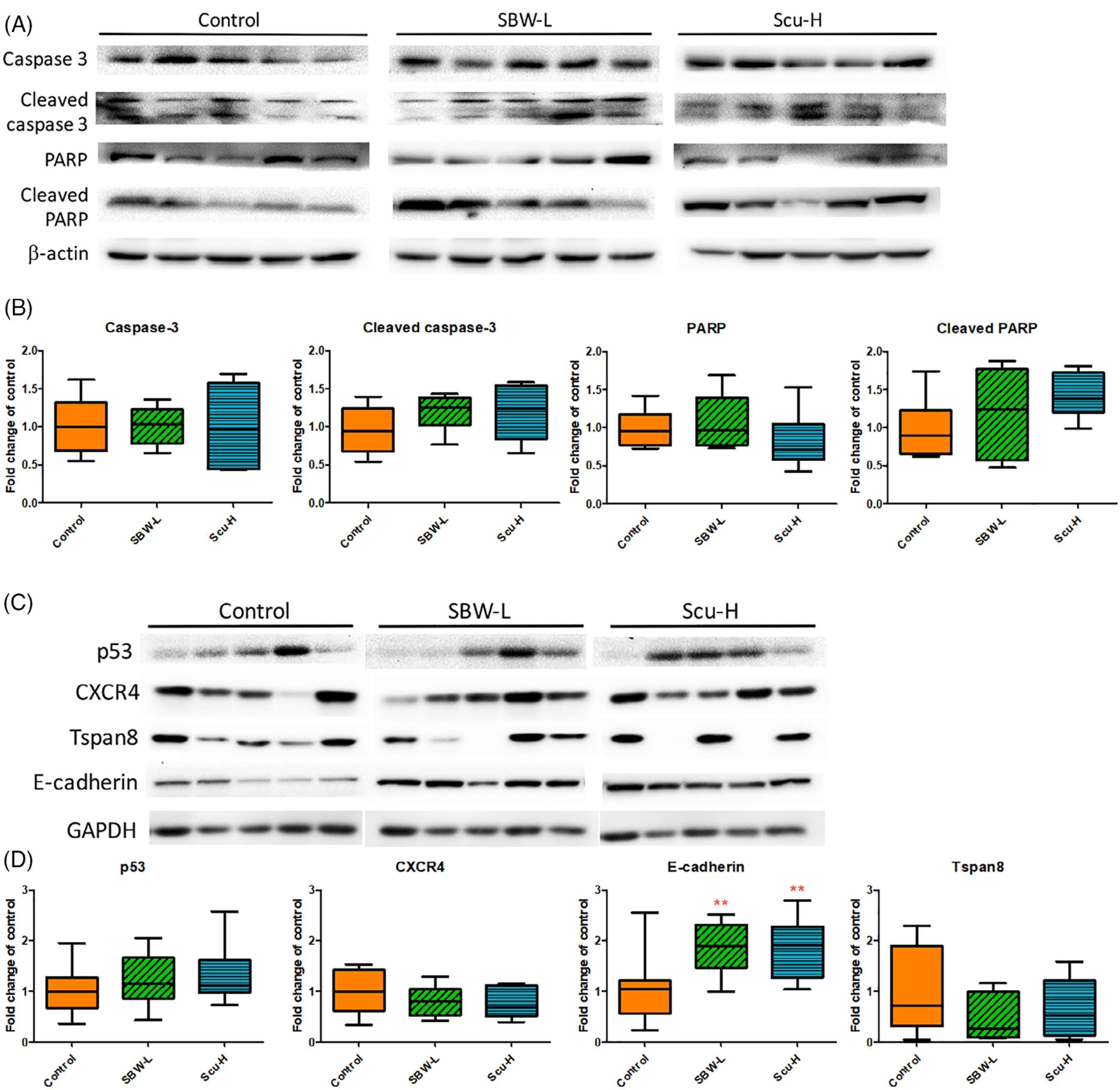

FIGURE 4 Effects of SB water extract (SBW) and scutellarin (Scu) treatments on protein expressions in colon tumors. Proteins were extracted from the tumors and subjected to western blotting analysis. Representative blots (5 protein samples from each group) showed the effects of SBW and Scu treatments on the expressions of (a) caspase 3 and PARP and their cleaved form and (c) growth- or metastasis-related molecules. (b, d) Histograms showed the quantitative results of the expressions of target proteins which were normalized with corresponding β-actin/GAPDH expressions and expressed as fold change of control. Data were expressed in box and whisker plots of 11–12 mice each group. Differences among treatment groups were determined by one-way ANOVA followed by Tukey's multiple comparisons test, $^{**}p < .01$ as compared with control group. ANOVA, analysis of variance [Colour figure can be viewed at wileyonlinelibrary.com]

metastasis related proteins, CXCR4, E-cadherin and Tspan 8 were determined in the tumor proteins. As shown in Figure 4C,D, the chemokine CXCR4, which was associated with high incidence of metastasis and poor prognosis of colon cancer patients (S. S. Zhang et al., 2012), was shown to have a trend of decrease after SBW-L or Scu-H treatments. Besides, Tspan 8 (a modifier of cancer cell motility [Greco et al., 2010]) expression was lower in SBW-L-treated mice tumors when compared to that in untreated control mice. In addition, the expressions of E-cadherin in tumors were significantly augmented in SBW-L- or Scu-H-treated mice ($p < .01$).

3.5 | SBW altered the expressions of ERK, β-catenin and Src in HCT116 tumors

In order to further elucidate the underlying mechanism that involved in the SBW-induced anti-tumor and anti-metastatic effects, the expressions of pERK, β-catenin and Src were determined using western blotting. As shown in Figure 5A,B, the expression levels of phosphorylated-ERK were reduced in tumors of SBW-L and Scu-H treatment groups. Besides, Src and β-catenin expressions were shown to be significantly suppressed in SBW-L- and Scu-H-treated groups, respectively. These data suggested that the mechanisms involved in the inhibitory effects of SBW and Scu would be different.

3.6 | SBW inhibited orthotopic colon tumor growth and metastasis in Colon26-Luc tumor-bearing mice

To further confirm the anti-tumor and anti-metastatic efficacy of SBW and Scu, mice bearing orthotopic colon tumors were received SBW-L and Scu-H treatments. Tumor growth in the rectum, which could not be observed nor measured during treatment, were monitored using bioluminescence imaging. As shown in Figure 6A, the signal emission (expressed as average radiance) increased from Day 9 onwards which indicated that an increase in tumor burden in the rectum. Treatment with SBW-L resulted in obvious inhibitory effect on tumor growth in rectum. On Day 18, significant decrease of tumor size was observed when compared with untreated control group ($p < .05$). However, Scu-H treatment did not effectively suppress the tumor growth till Day 18 (Figure 6B).

At the end of experiment, lung metastasis was also evaluated by bioluminescence imaging. As shown in Figure 6C, the signal emission of lungs in untreated control group was the highest, indicating that the luciferase-tagged tumor cells migrated to the lungs and proliferated rapidly. Colon26-Luc tumor burden in terms of signal emission of bioluminescent tumor cells in lungs was decreased by 94.0 and 61.5% in SBW-L and Scu-H-treated groups, respectively. Results suggested that treatment with SBW-L and Scu-H could decrease the lung metastasis in orthotopic colon tumor-bearing mice, although the signal emission differences among groups were not statistically significant.

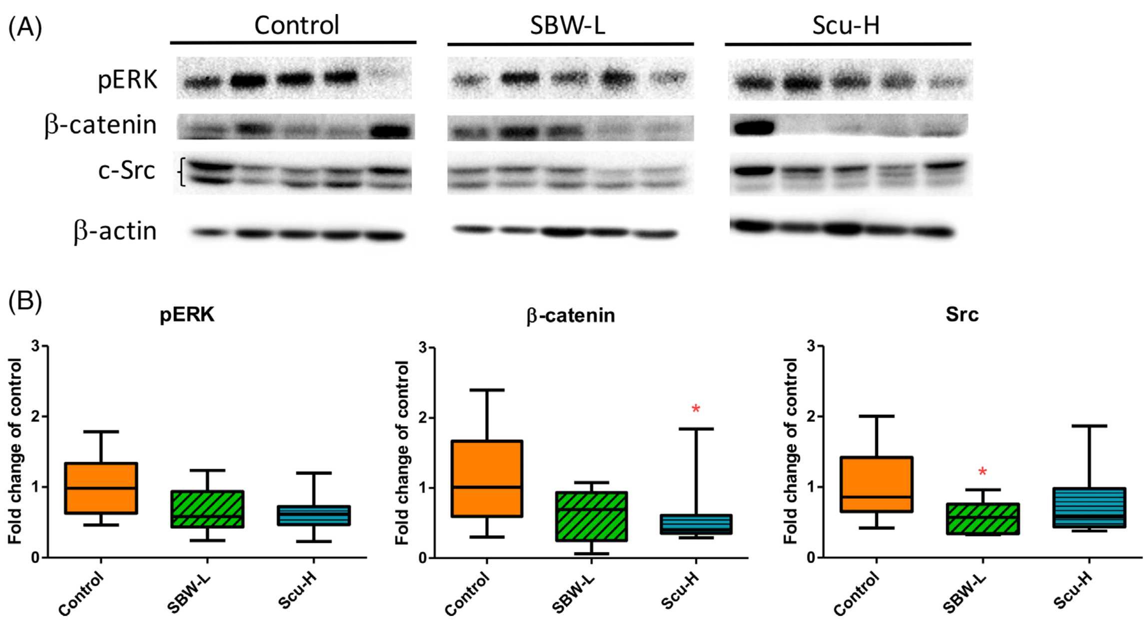

FIGURE 5 Expressions of pERK, β-catenin and Src in tumors of SBW- and Scu-treated mice. Proteins were extracted from the tumors and subjected to western blotting analysis. (a) Representative blots (5 protein samples from each group) showed the effects of SBW and Scu treatments on the expressions of pERK, β-catenin and Src. (b) Histograms showed the quantitative results of the expressions of target proteins which were normalized with corresponding β-actin expressions and expressed as fold change of control. Data were expressed in box and whisker plots of 11–12 mice each group. Differences among treatment groups were determined by one-way ANOVA followed by Tukey's multiple comparisons test, *$p < .05$ as compared with control group. ANOVA, analysis of variance; SBW, SB water extract [Colour figure can be viewed at wileyonlinelibrary.com]

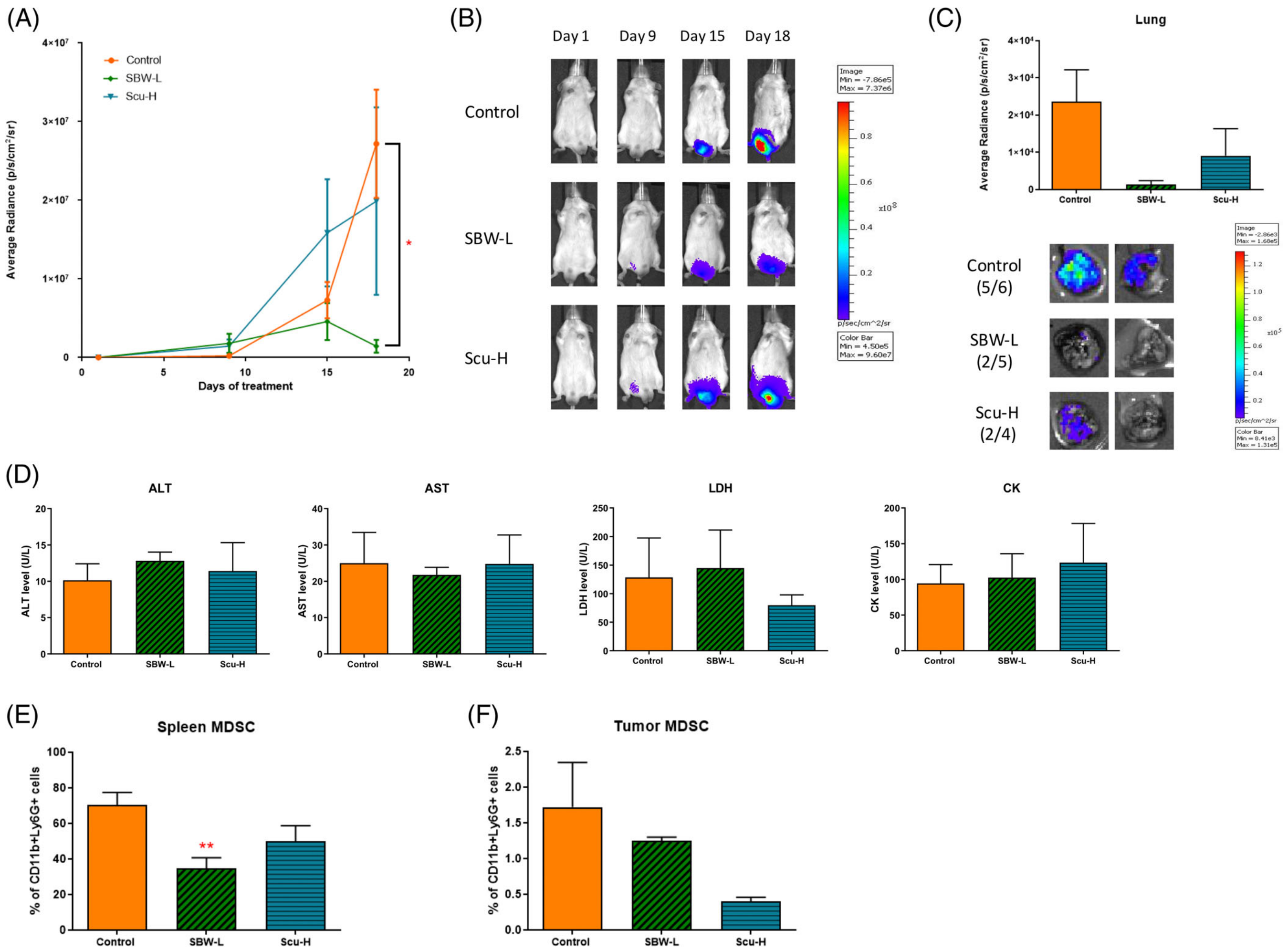

FIGURE 6 Tumor burden change during SBW and Scu treatments in orthotopic colon tumor-bearing mice. (a) Graph showed the bioluminescence measurements according to the average radiance. (b) Representative images of different groups at each time point of IVIS scan. (c) Graph showed the bioluminescence measurements (expressed as average radiance) in lungs. Representative images of lungs obtained from different groups at end point of IVIS scan, and the fractional number revealed the frequency of metastasis. (d) Effects of SBW and Scu treatments on the levels of plasma enzymes ALT, AST, LDH and CK in HCT116 tumor-bearing nude mice after 28 days of treatment. Spleens and tumors were collected from SBW- or Scu-treated or untreated control mice. Populations of MDSC in (e) spleens and (f) tumors were determined by flow cytometry after staining with PE-anti-mouse CD 11b and FITC-anti-mouse Ly-6G (Gr-1) antibodies. Data were expressed as mean ± *SEM* in (a) or (c, d, e, f), *n* = 7. Differences among treatment groups were determined by one-way ANOVA followed by Tukey's multiple comparisons test, **p* < .05, ***p* < .01 as compared with control group. ANOVA, analysis of variance; SBW, SB water extract [Colour figure can be viewed at wileyonlinelibrary.com]

The levels of liver-specific plasma enzymes ALT, AST and LDH and muscle-specific enzyme CK in Colon26-Luc tumor-bearing mice were also determined and results showed that the no significant difference among treatment groups (Figure 6D), suggesting that SBW and Scu treatments did not cause toxic effects in immune competent mice.

On the other hand, the MDSC populations were determined in spleens and tumors of mice in treated or untreated groups. Results showed that the MDSC population in spleens of mice treated with SBW-L was significantly decreased (*p* < .01, Figure 6E). While the MDSC population in tumors was reduced in Scu-H treated mice, although the changes in treated groups were not significantly different from the untreated control group (*p* > .05, Figure 6F).

4 | DISCUSSION

The herb SB is commonly used as herbal medicines for treating cancer (Chen et al., 2018). Despite the high frequency of use in clinics, there are only a few preclinical studies on the anti-tumor effects of SBW (Dai et al., 2011; Wong et al., 2009). The dosages of SBW used in the previous animal studies (including liver and prostate cancer) were in fact higher than the maximum human equivalent dose recommended in CP 2015 (i.e., 1,230 mg/kg). On the other hand, the promising anti-tumor activities of solvent (ethanol/ methanol/ chloroform) extracts of SB have been intensively reported in the last decade. Recent studies demonstrated the underlying mechanisms of solvent (ethanol/ chloroform) extracts of SB in colorectal cancer cells, which involved the IL-6-inducible STAT3 pathway (Q. Jiang et al., 2015), PI3K/AKT and TGF-β/Smad signaling pathways (Jin et al., 2017; J. Lin, Feng, et al., 2017), as well as the miR-34a (L. Zhang et al., 2017). Nonetheless, the efficacy of SBW at clinical relevant dosages in colon tumor-bearing mice has never been investigated.

In our in vitro study, SBW and Scu, a flavonoid found in SB and also serve as the chemical marker of SB as listed in CP 2015, were initially evaluated for their cytotoxicities in two human colon cell lines, HCT116 and HT-29. This was because inhibitory effects of SB ethanol extract or Scu were shown in these cell lines in previous studies (L. Wei et al., 2012; Yang et al., 2017). However, HCT116 cells were shown to be more sensitive to SBW and Scu treatment in our study (data not shown). Hence, HCT116 cells were selected for the subsequent experiments. The IC_{50} value of Scu in HCT116 cells obtained here (105.4 μM) was similar to those reported previously (Ni, Tang, Li, He, & Rao, 2018; Yang et al., 2017), except one recent study in which the IC_{50} values of Scu in four colon cancer cell lines were below 10 μM (Zhu, Mao, Liu, Tao, & Yan, 2017). On the other hand, both SBW and Scu at non-cytotoxic concentrations were shown to inhibit migration of colon cancer cells. The results of Scu were consistent with previous studies using other cancer cells (Ke et al., 2017; H. Li et al., 2010; Zhu et al., 2017). Besides, the inhibition of SBW (200 μg/ml, which contain 2.3 μM Scu) on migration reached almost 50%, while the inhibitory activities of Scu alone would be observed only at greater than 10 μM (data not shown). It is possible that other components, such as scutellarein (Shi et al., 2015), flavonoids (Zheng, Kang, Liu, & Guo, 2018) and water soluble polysaccharide (H. Li et al., 2019), in SBW contribute to the anti-migratory activities.

This study also aimed to validate the in vivo efficacy of SBW and Scu at the CP 2015 recommended human equivalent dosages, which were not usually considered in other preclinical animal studies. In fact, our results showed the significant anti-tumor effects of SBW (at both low and high dosages) and Scu (at the high dosage), and the efficacies of SBW and Scu could also be compared in the same set of experiments. The inhibitory effects of SBW-L and Scu-H on tumor growth and lung metastasis were found to be comparable, suggesting that other active components present in SBW would play roles on anti-tumor and anti-metastasis activities in colon tumor-bearing mice. For example, scutellarein, which is the parent compound of Scu before glucuronidation of its 7-OH group, was shown to induce apoptosis in lung cancer cells (Cheng, Hu, Yang, Lee, & Kao, 2014), colon cancer cells (Guo, Yang, & Zhu, 2018; F. Wang et al., 2014) and attenuate fibrosarcoma in vivo (Shi et al., 2015). Although the abundance of scutellarein was half of Scu in SB (Peng et al., 2015), the cytotoxicity of scutellarein was higher (about 5 times) than Scu in colon cancer cells (F. Wang et al., 2014). Besides, water soluble polysaccharides that exhibit apoptotic and anti-proliferative properties (H. Li et al., 2019; Sun, Sun, & Wang, 2017) could also contribute to the anti-tumor activities of SBW. The mixtures of these components in SBW would exert additive/synergistic activities in tumor-bearing mice.

Apart from the changes on tumor weights and lung metastasis levels, the expressions of apoptosis- and metastasis-related proteins in the tumors were determined so that the underlying actions of SBW and Scu could be revealed. Previous studies demonstrated that SB ethanol extract (2 g/kg, intragastric administration) suppressed angiogenesis via Hedgehog pathway (L. Wei et al., 2012) and inhibited the activation of STAT3, Erk, p38, Wnt/β-catenin signaling pathways in colon HT29 tumors (J. Lin et al., 2014; L. H. Wei et al., 2017). A recent study also reported that orally administered Scu (10 mg/kg) decreased Ki67- and CD34-positive cells in colon HCT116 tumors and lung metastasis number (Zhu et al., 2017). Nonetheless, the increased expressions of E-cadherin in colon tumors after SBW (615 mg/kg) or Scu (7 mg/kg) treatments were shown here. E-cadherin was known to correlate with metastasis in colon cancer (Park et al., 2016; B. Yue et al., 2016) and could be up-regulated by Scu in tongue cancer cells (H. Li et al., 2010). Changes of E-cadherin expression in tumors might be responsible for the decreased lung metastasis in mice after SBW or Scu treatments. In addition, the chemokine receptor CXCR4 and tetraspanin Tspan 8, which are closely related to colon cancer metastasis (Huang et al., 2012; S. S. Zhang et al., 2012), were shown for the first time that their expressions in colon tumors could be down-regulated after SBW and Scu treatments. Furthermore, the expressions of pERK and β-catenin were also down-regulated by SBW and Scu treatments. These results are in consistent with previous studies using SB ethanol extract (2 g/kg) in colon tumor-bearing mice (J. Lin et al., 2014; L. H. Wei et al., 2017). Interestingly, our SBW treatment induced apparent down-regulation of Src in colon tumors. Src was known to enhance metastasis by altering alpha3 integrin expression in colon cancer cells (Kline, Olson, & Irby, 2009) and was recently regarded as an important molecular markers for prediction of prognosis in Stage IV colorectal cancer (Shimada et al., 2019). The potential of SBW on regulating Src expression or Src kinase activity in colon cancer warrants further investigation. Nevertheless, the target proteins expressions in tumors varied as shown in Figures 4 and 5. It is possible that tumor sizes would affect the bioavailability of SBW active components and scutellarin within the tumors. Since the SBW and scutellarin were orally administered, they or their metabolites could be transported to the tumors via blood vessels, whose numbers were varied in each tumors, the availability of the anti-tumor components would be different in each tumors. Hence, the responses (cancer cell death, blood vessel reduction, etc.) of tumor cells would be diverse, and the expressions of growth- and metastasis-related proteins would be diverse. Despite the probable variation, the target proteins expressions were determined in 11–12 tumors from different treatment groups and the data summarized in Figures 4 and 5 could show some significant changes after SBW or scutellarin treatments. On the other hand, the results from the bioluminescence orthotopic colon tumor model further confirmed the anti-tumor as well as anti-metastatic efficacy of SBW at its clinical relevant dosages in immunocompetent mice. The significant inhibition in tumor burden in rectum and metastasis to lungs by SBW was demonstrated for the first time using IVIS. The sharply reduced lung metastasis was further confirmed by H&E staining (data not shown). Besides, both SBW-L and Scu-H could decrease the population of MDSC in colon tumor-bearing mice. MDSC have been regarded as promotor of tumor progression through multiple mechanisms in colorectal cancer (Y. Wang et al., 2019). Hence, the modulatory effects of SBW and Scu in the microenvironment of colon tumor-bearing mice are worth further investigation.

In conclusion, the present study demonstrated the inhibitory activities of SBW on colon cancer cells growth and migration in vitro, as well as the tumor growth and metastasis in colon tumor-bearing mice. In particular, the regulation on the metastasis-related proteins in tumors, which was induced by the 4-weeks SBW treatment, was first reported. Our findings provide evidence that SBW (at the mouse equivalent dosages to clinical dosages recommended by Chinese Pharmacopoeia) could exert anti-tumor and anti-metastatic effects in colon cancer animal model.

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CONFLICT OF INTEREST

The authors declare no potential conflict of interest.

AUTHOR CONTRIBUTIONS

Grace Gar-Lee Yue, **Clara Bik-San Lau** and **Yuk-Yu Chan:** Conceived and designed the experiments. **Yuk-Yu Chan**, **Wenjing Liu**, **Si Gao**, **Grace Gar-Lee Yue**, **Julia Kin-Ming Lee** and **Kit-Man Lau:** Conducted experiments. **Grace Gar-Lee Yue**, **Si Gao** and **Yuk-Yu Chan:** Performed data analysis. **Chun-Wai Wong:** Contributed in chemical analysis. **Grace Gar-Lee Yue** and **Yuk-Yu Chan:** Wrote the paper that was revised by **Clara Bik-San Lau**. All authors read and approved the final manuscript.

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